Recommendations — gbm-mtap-cdkn2a-idhwt-subtotal-c4bq

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-08-12

Recommendations — gbm-mtap-cdkn2a-idhwt-subtotal-c4bq

Profile snapshot

- **Primary site:** brain
- **Histology:** glioblastoma, IDH-wildtype, WHO grade 4 (CNS WHO 2021); presenting tumor demonstrated intratumoral hemorrhage
- **Stage:** newly diagnosed; status post subtotal (partial) resection — gross-total resection NOT achieved; residual enhancing disease presumed on post-op MRI; no extracranial disease (GBM does not stage by AJCC TNM)
- **ECOG:** 1
- **Age band:** 60-69
- **Sex:** female
- **Biomarkers:** IDH1 wild-type; CDKN2A (p16) homozygous loss; MTAP loss (homozygous deletion of 9p21, co-deleted with CDKN2A); MGMT promoter methylation unknown (unknown); EGFR unknown (amplification / EGFRvIII status not stated) (unknown); TERT promoter unknown (unknown); Chromosome 7 gain / chromosome 10 loss (+7/-10) unknown (unknown); BRAF V600E unknown (unknown); NTRK fusions unknown (unknown); MMR / MSI unknown (unknown)
- **Efficacy/toxicity weight:** 0.7

12 ranked therapeutic options across 6 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Idhwt Gbm Backbone interventions

1. Stupp regimen + TTFields: RT 60 Gy/30 fx + concurrent TMZ 75 mg/m^2/day × 6 weeks, then 6 cycles adjuvant TMZ 150-200 mg/m^2 d1-5 q28d, with Optune Gio (200 kHz) added from adjuvant cycle 1 at ≥18 hr/day

Likelihood of effect: High — Stupp HR_OS 0.63 (95% CI 0.52-0.75, replicated 5-yr) + EF-14 HR_OS 0.63 (95% CI 0.53-0.76) layered on TMZ; the 20.9-mo median is the threshold every other rank is measured against.

Toxicity burden: Moderate (G≥3 hematologic 14% adjuvant-phase, lymphopenia under TMZ requiring PJP prophylaxis, scalp dermatitis 52% any grade / 2% G≥3 under arrays)

Counter-productive MoA:Low (TMZ-induced hypermutation at recurrence can paradoxically open MMR-deficient ICI eligibility — a delayed pro-mechanism effect rather than a true counter-productive one.)

Overall: The OS-anchored standard the entire ranking is built around; every other rank either layers on this backbone or is measured against it.**

Key references: PMID 15758010 · PMID 15758009 · PMID 19269895 · PMID 29392280 · NCT00006353 · NCT00916409

2. GBM AGILE adaptive platform (NCT03970447) — Bayesian-randomized multi-site

backbone alternative; current arms AZD1390 + RT (newly diagnosed only), tinostamustine, ADI-PEG 20, troriluzole, VT1021

Likelihood of effect: Moderate — Bayesian randomization against SoC; current arms (AZD1390+RT, tinostamustine, ADI-PEG 20) are not MTAP-matched, so mechanism-axis upside is lower than BGB-58067.

Toxicity burden: Moderate (varies by arm assignment; AZD1390+RT carries CNS toxicity and lymphopenia signal, tinostamustine carries hematologic AE typical of alkylators)

Counter-productive MoA:Low (Arm randomization can land the patient on a non-mechanism-matched investigational agent — efficiency cost rather than counter-productive mechanism.)

Overall: **The strongest multi-site randomized-platform alternative when the Phoenix slot is infeasible; mechanism axis is weaker than BGB-58067 because no current arm targets MTAP.**

Key references: NCT03970447

3. SurVaxM (survivin peptide vaccine SVN53-67/M57-KLH) + GM-CSF on the SURVIVE phase 2b NCT05163080, layered on adjuvant TMZ

Likelihood of effect: Low-to-moderate — Ahluwalia 2023 single-arm n=63 reported PFS6 95.2%, mOS 25.9 mo against ~16-mo historical control; SURVIVE phase 2b is the randomized validation gate.

Toxicity burden: Low (injection-site reactions, low-grade flu-like symptoms; zero SurVaxM-attributable G≥3 SAEs in n=63)

Counter-productive MoA:Low (Vaccine adjuvant immune activation has no established counter-productive mechanism in GBM; the main concern is Optune incompatibility (foregoing EF-14-anchored OS) rather than a mechanism-level risk.)

Overall: **The lowest-toxicity add-on in the dossier with a real but unreplicated single-arm signal; forces a binary choice against TTFields because SURVIVE excludes concurrent Optune.**

Key references: PMID 36473145 · NCT05163080

4. CAN-3110 (rQNestin34.5v.2) oncolytic HSV-1 intratumoral injection — recurrence pre-position; order HSV-1 IgG serology now to inform the prior probability of benefit

Likelihood of effect: Low overall, moderate in HSV-1 seropositive subset — Ling 2023 single-dose mOS 11.6 mo overall, 14.2 mo seropositive vs ~9 mo historical recurrent-GBM benchmark.

Toxicity burden: Low (no DLTs across the Ling 2023 dose range; perilesional inflammation is the modal AE)

Counter-productive MoA:Low (Perilesional inflammation from viral lysis could re-bleed in a pre-bled brain — a hemorrhage-history-specific relative caution rather than a class-level counter-productive mechanism.)

Overall: **Recurrence-time pre-position with a cheap up-front serology that doubles the prior probability of benefit; hemorrhage history is a relative caution not an exclusion.**

Key references: PMID 37853118 · NCT03152318

5. FUS-mediated BBB opening (SonoCloud-9 / SONOBIRD NCT04528680) at first recurrence — pre-position at a US SONOBIRD site (Northwestern/Sonabend, Columbia, MD Anderson)

Likelihood of effect: Low-to-moderate — Sonabend 2023 (n=17) demonstrated 4-6× peri-tumoral drug concentration enhancement; SONOBIRD OS data not yet mature.

Toxicity burden: Low (no implant-related deaths in Sonabend 2023; tolerated through 6 cycles)

Counter-productive MoA:Low (BBB opening could theoretically increase off-target CNS exposure of partner cytotoxics, but the Sonabend cohort showed no implant-related deaths or severe CNS toxicity.)

Overall: Recurrence-time delivery-enhancement pre-position; requires pre-screening at a SONOBIRD US site at the first-recurrence resection conversation.****

Key references: PMID 37087102 · NCT04528680

Evidence in detail

Stupp regimen + TTFields: RT 60 Gy/30 fx + concurrent TMZ 75 mg/m^2/day × 6 weeks, then 6 cycles adjuvant TMZ 150-200 mg/m^2 d1-5 q28d, with Optune Gio (200 kHz) added from adjuvant cycle 1 at ≥18 hr/day

Stupp R et al. *N Engl J Med* 2005

- **Disease context:Newly diagnosed glioblastoma** (1L) — Adults age 18-70 with newly diagnosed histologically confirmed GBM, WHO PS 0-2, post-resection or biopsy. MGMT status not used for stratification at randomization but later shown to be the strongest effect modifier.
- **n:** 573
- **Toxicities:grade 3-4 hematologic AE during concurrent phase** (grade >=3): 7% (n=284); **grade 3-4 hematologic AE during adjuvant phase** (grade >=3): 14% (n=223); **any grade 3-4 AE** (grade >=3); **pneumocystis pneumonia** (grade any)
- **Efficacy:HR_OS** 0.63 (95% CI 0.52-0.75); **OS** 14.6 months; **OS_rate** 26.5%

Hegi ME et al. *N Engl J Med* 2005

- **Disease context:Newly diagnosed glioblastoma; MGMT-methylated vs unmethylated subsets** (1L) — MGMT promoter methylation by methylation-specific PCR on 206 of 573 EORTC 26981 patients with assessable tumor. Methylated 45%, unmethylated 55%.
- **n:** 206
- **Toxicities:** No new safety signals; this is a biomarker analysis of EORTC 26981, safety reported in the parent trial.
- **Efficacy:OS** 21.7 months; **OS** 15.3 months; **OS** 12.7 months

Stupp R et al. *Lancet Oncology* 2009

- **Disease context:Newly diagnosed glioblastoma — long-term follow-up** (1L) — Full EORTC 26981 cohort with 5+ years of follow-up.

- **n:** 573
- **Toxicities:** No new late toxicities beyond the parent trial. MGMT methylation was the dominant predictor of long-term survival; methylated MGMT and TMZ exposure together identified the patients who reached 5-year survival.
- **Efficacy:OS_rate** 27.2%; **OS_rate** 16.0%; **OS_rate** 9.8%; **HR_OS** 0.6 (95% CI 0.5-0.7)

Stupp R et al. *JAMA* 2017

- **Disease context:Newly diagnosed glioblastoma post-RT+TMZ; adjuvant maintenance** (maintenance) — Age 18+, completed RT+TMZ without progression, KPS >=70, both MGMT methylated and unmethylated enrolled.
- **n:** 695
- **Toxicities:scalp dermatitis / irritation under arrays** (grade any): 52% (n=466); **scalp dermatitis** (grade >=3): 2% (n=466); **thrombocytopenia** (grade >=3): 4% (n=466); **fatigue** (grade any)
- **Efficacy:HR_OS** 0.63 (95% CI 0.53-0.76); **OS** 20.9 months; **PFS** 6.7 months; **OS_rate** 43%; **OS_rate** 13%

SurVaxM (survivin peptide vaccine SVN53-67/M57-KLH) + GM-CSF on the SURVIVE phase 2b NCT05163080, layered on adjuvant TMZ

Ahluwalia MS et al. *Journal of Clinical Oncology* 2023

- **Disease context:Newly diagnosed glioblastoma** (1L (adjuvant)) — 63 adults with newly diagnosed GBM who completed Stupp RT+TMZ without progression; KPS >=70; survivin expression confirmed by IHC.
- **n:** 63
- **Toxicities:injection-site reaction** (grade any); **fatigue** (grade any); **any SAE attributable to SurVaxM** (grade >=3): 0% (n=63)
- **Efficacy:PFS_rate** 95.2%; **PFS** 11.4 months; **OS** 25.9 months; **OS_rate** 51%; **OS_rate** 41%

CAN-3110 (rQNestin34.5v.2) oncolytic HSV-1 intratumoral injection — recurrence pre-position; order HSV-1 IgG serology now to inform the prior probability of benefit

Ling AL et al. *Nature* 2023

- **Disease context:Recurrent high-grade glioma (mostly recurrent GBM)** (2L+) — 41 evaluable patients with recurrent HGG, KPS >=70, accessible enhancing tumor for stereotactic injection. Pre-existing HSV-1 serology was the key effect-modifier.
- **n:** 41
- **Toxicities:any DLT** (grade >=3): 0% (n=41); **perilesional edema** (grade any); **headache** (grade any); **seizure** (grade any); **intratumoral hemorrhage** (grade any)
- **Efficacy:OS** 11.6 months (95% CI 7.8-14.9); **OS** 14.2 months; **OS** 7.8 months; **biomarker**

FUS-mediated BBB opening (SonoCloud-9 / SONOBIRD NCT04528680) at first recurrence — pre-position at a US SONOBIRD site (Northwestern/Sonabend, Columbia, MD Anderson)

Sonabend AM et al. *Lancet Oncology* 2023

- **Disease context:Recurrent glioblastoma — peri-tumoral BBB disruption to enable systemic chemotherapy delivery** (2L+) — 17 adults with recurrent GBM, KPS >=60, surgically implanted with the SonoCloud-9 skull device at time of resection. Single-center Northwestern.
- **n:** 17

- **Toxicities:implant-related complication** (grade >=3): 0% (n=17); **FUS-related transient edema** (grade any); **nab-paclitaxel related peripheral neuropathy** (grade >=3)
- **Efficacy:biomarker**; **DCR**; **TRAE_rate**

Mtap Loss interventions

1. BGB-58067 (brain-penetrant MTA-cooperative PRMT5 inhibitor) on the Sanai phase 0/2 NCT07485049 at Ivy Brain Tumor Center, Phoenix — Arm A (unmethylated MGMT + RT) or Arm B (methylated MGMT + RT + TMZ); layered on the Stupp backbone, not substituting for it

Likelihood of effect: Moderate uncertainty — mechanism-matched at MTAP/CDKN2A with brain-penetrant chemistry but no published BGB-58067 efficacy on the molecule itself; class proxy ORR ~11% in pre-treated all-comer MTAP-null tumors (anvumetostat MTAPESTRY-101).

Toxicity burden: Low (class proxy: G≥3 TRAE 13.8%, discontinuation 2.5%, no treatment-related deaths in anvumetostat MTAPESTRY-101; BGB-58067 CNS-specific tox unpublished)

Counter-productive MoA:Moderate (Inadequate CNS exposure was the proximate failure mode of TNG908 — design rationale addresses it but human CSF PK on BGB-58067 itself is not yet public.)

Overall: The cleanest mechanistic match in the dossier for this patient's 9p21 co-deletion; biomarker fit is exact and the trial preserves the Stupp backbone, but the molecule has no published human efficacy and the class's CNS record is two failures.****

Key references: NCT07485049 · doi:10.1016/j.annonc.2024.06.005 · PMID 39282709

2. Navlimetostat (BMS-986504), MTA-cooperative brain-penetrant PRMT5 inhibitor — dedicated recurrent-GBM CNS-penetration trial NCT06883747 (Sanai/Ivy, recruiting now) plus the MountainTAP-5 basket NCT07492680 (newly-diagnosed GBM + TMZ + RT cohort, opens July 2026); the same molecule showed target engagement in GBM tumor tissue and cross-tumor NSCLC ORR 29%

Likelihood of effect: Moderate — human GBM target engagement now shown (SNO 2025: 77% SDMA reduction, unbound drug above CNS threshold) plus cross-tumor NSCLC ORR 29% / mDoR 10.5 mo; GBM efficacy on the molecule unpublished.

Toxicity burden: Low (cross-tumor: G≥3 TRAE 14%, no treatment-related deaths, grade 1-3 anemia/neutropenia/thrombocytopenia; GBM-specific AE table not yet reported)

Counter-productive MoA:Low (The class CNS-exposure failure mode (TNG908) is now directly addressed — GBM tumor unbound drug exceeded the 5x-IC50 penetration threshold on the SNO 2025 readout.)

Overall: Post-board addition on the MTAP axis with human GBM proof-of-mechanism (CNS penetration + target engagement) and cross-tumor NSCLC activity, but no GBM efficacy yet and no board vote; actionable-now GBM slot is recurrence-gated.****

Key references: NCT06883747 · NCT07492680 · doi:10.1093/neuonc/noaf201.1669 · doi:10.1016/j.jtho.2025.09.054 · PMID 39282709

3. TNG456 (brain-penetrant MTA-cooperative PRMT5 inhibitor) +/- abemaciclib on NCT06810544 (Tango, recruiting) — the single most mechanism-complete row for this case: TNG456 hits the MTAP-loss PRMT5 axis and the abemaciclib arm hits the CDKN2A-loss CDK4/6 axis, both of the patient's nominated 9p21 targets at once; recurrence-gated

Likelihood of effect: Moderate mechanistically, unproven clinically — the only dual-axis (MTAP PRMT5 + CDKN2A CDK4/6) row; evidence is preclinical (Briggs SNO 2025: 55x MTAP-null selectivity, xenograft regression) plus a trial-in-progress.

Toxicity burden: Moderate (no human GBM AE table yet; abemaciclib arm contributes G≥3 diarrhea, neutropenia, fatigue; PRMT5i + CDK4/6i additive myelosuppression unquantified)

Counter-productive MoA:Moderate (TNG908's inadequate-CNS-exposure failure is the class caution; TNG456 is the engineered successor but human CNS PK is not yet public.)

Overall: The most mechanism-complete option in the dossier — hits both the MTAP and CDKN2A 9p21 axes at once — but evidence is preclinical plus a trial-in-progress, it is recurrence-gated, and it has no board vote.****

Key references: NCT06810544 · doi:10.1093/neuonc/noaf201.0647 · PMID 39282709

Evidence in detail

BGB-58067 (brain-penetrant MTA-cooperative PRMT5 inhibitor) on the Sanai phase 0/2 NCT07485049 at Ivy Brain Tumor Center, Phoenix — Arm A (unmethylated MGMT + RT) or Arm B (methylated MGMT + RT + TMZ); layered on the Stupp backbone, not substituting for it

Rodon J et al. *Cancer Discovery* 2025

- **Disease context:MTAP-deleted advanced solid tumors** (2L+) — Combined preclinical + early clinical translational characterization; patient cohorts overlap with the AnnOnc 2024 phase 1 paper.
- **n:** 80
- **Toxicities:** Safety detailed in the AnnOnc 2024 phase 1 paper; same dataset. Translational sections in this paper carry preclinical hematologic-sparing data (no impact on normal hematopoietic lineages in xenografts).
- **Efficacy:biomarker**; **ORR**

Navlimetostat (BMS-986504), MTA-cooperative brain-penetrant PRMT5 inhibitor — dedicated recurrent-GBM CNS-penetration trial NCT06883747 (Sanai/Ivy, recruiting now) plus the MountainTAP-5 basket NCT07492680 (newly-diagnosed GBM + TMZ + RT cohort, opens July 2026); the same molecule showed target engagement in GBM tumor tissue and cross-tumor NSCLC ORR 29%

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- **Efficacy:biomarker**; **ORR**

Mgmt Methylated interventions

1. CeTeG / NOA-09: lomustine (CCNU) 100 mg/m^2 day 1 + TMZ 100-200 mg/m^2 days 2-6 every 6 weeks × 6 cycles, with concurrent RT 60 Gy — conditional on MGMT promoter methylated AND adequate baseline marrow reserve *(conditional on mgmt_methylated positive)*

Likelihood of effect: High in MGMT-methylated subset — Herrlinger 2019 HR_OS 0.60 (95% CI 0.35-1.03), mOS 48.1 vs 31.4 mo; CI precision is modest at n=141 and unreplicated.

Toxicity burden: Moderate (G≥3 hematologic 41%: lymphopenia 27%, thrombocytopenia 16%, neutropenia 13%; no treatment-related deaths in CeTeG arm)

Counter-productive MoA:Low (Cumulative CCNU myelosuppression can foreclose later cytotoxic combinations at recurrence (lomustine + bevacizumab) — a sequencing rather than a counter-productive concern.)

Overall: The largest published front-line OS signal in MGMT-methylated GBM, conditional on a result the workup row will return; toxicity at age 66 is the binding constraint and the CI is wider than the headline number suggests.****

Key references: PMID 31488360 · NCT01149109

Evidence in detail

CeTeG / NOA-09: lomustine (CCNU) 100 mg/m^2 day 1 + TMZ 100-200 mg/m^2 days 2-6 every 6 weeks × 6 cycles, with concurrent RT 60 Gy — conditional on MGMT promoter methylated AND adequate baseline marrow reserve

Herrlinger U et al. *Lancet* 2019

- **Disease context:Newly diagnosed MGMT-methylated glioblastoma** (1L) — Adults age 18-70, KPS >=70, newly diagnosed GBM with MGMT promoter methylated by MSP, post-resection or biopsy. 17 German university centers.
- **n:** 141
- **Toxicities:any grade 3-4 hematologic AE** (grade >=3): 41% (n=63); **lymphopenia** (grade >=3): 27% (n=63); **thrombocytopenia** (grade >=3): 16% (n=63); **neutropenia** (grade >=3): 13% (n=63); **any grade 3-4 hematologic AE** (grade >=3): 13% (n=66)
- **Efficacy:HR_OS** 0.6 (95% CI 0.35-1.03); **OS** 48.1 months

Braf V600E interventions

1. Tumor-agnostic conditional plays unlocked by the rank-1 NGS panel: dabrafenib + trametinib for BRAF V600E (ROAR HGG cohort); larotrectinib or entrectinib for NTRK fusion-positive; pembrolizumab for MSI-H / dMMR or TMB-H — all recurrence-only and biomarker-conditional

Likelihood of effect: Low prior probability across all three biomarkers in adult IDH-WT GBM (BRAF V600E ~1-2%, NTRK <1%, MMR-D uncommon); ORR conditional on a positive hit is ROAR 33%, larotrectinib 75-80%, pembrolizumab TMB-H 29%.

Toxicity burden: Moderate (class AEs by axis; mostly manageable on outpatient monitoring)

Counter-productive MoA:Low (Each axis is biomarker-selected; mechanism-level risk only arises if a positive call is acted on without confirmation (e.g. NTRK call without RNA-fusion verification).)

Overall: Free conditional options that fall out of the rank-1 NGS bundle at zero marginal cost; low prior probability across all three but the upside on a positive hit reframes the recurrence-time decision tree.****

Key references: PMID 34838156 · PMID 29466156 · PMID 32919526 · NCT02034110 · NCT02122913 · NCT02628067

Evidence in detail

Tumor-agnostic conditional plays unlocked by the rank-1 NGS panel: dabrafenib + trametinib for BRAF V600E (ROAR HGG cohort); larotrectinib or entrectinib for NTRK fusion-positive; pembrolizumab for MSI-H / dMMR or TMB-H — all recurrence-only and biomarker-conditional

Wen PY et al. *Lancet Oncology* 2022

- **Disease context:BRAF V600E-mutant high-grade glioma (HGG) — recurrent after standard therapy** (2L+) — Adults with BRAF V600E-mutant HGG (glioblastoma, anaplastic astrocytoma, anaplastic ganglioglioma, anaplastic pleomorphic xanthoastrocytoma), KPS >=70, prior RT and standard chemotherapy. 13 of 45 HGG patients had GBM.
- **n:** 45
- **Toxicities:any grade >=3 AE** (grade >=3): 53% (n=58); **fatigue** (grade >=3): 9% (n=58); **neutrophil count**

decreased (grade >=3): 9% (n=58); **headache** (grade >=3): 5% (n=58); **neutropenia** (grade >=3): 5% (n=58); **pyrexia** (grade any); **rash / dermatitis** (grade any)
- **Efficacy:ORR** 33% (95% CI 20-49); **DoR** 13.6 months; **PFS** 4.5 months; **OS** 17.6 months; **OS_rate** 41%

Drilon A et al. *N Engl J Med* 2018

- **Disease context:TRK fusion-positive advanced solid tumors (tumor-agnostic)** (2L+) — 55 patients (children + adults) with TRK fusion-positive tumors pooled from three early-phase trials. 17 tumor types represented; primary CNS not in this initial pool.
- **n:** 55
- **Toxicities:any grade >=3 AE** (grade >=3); **anemia** (grade >=3); **increased AST** (grade >=3): 3% (n=55); **increased ALT** (grade >=3): 3% (n=55); **dizziness** (grade any): 22% (n=55)
- **Efficacy:ORR** 75% (95% CI 61-85); **ORR** 80%; **DoR**; **CR_rate** 13%

Marabelle A et al. *Lancet Oncology* 2020

- **Disease context:TMB-H (>=10 mut/Mb) advanced solid tumors, tumor-agnostic** (2L+) — 102 patients with TMB-H by FoundationOne CDx from 10 KEYNOTE-158 tumor-type cohorts. Glioma not specifically in the primary TMB-H cohort.
- **n:** 102
- **Toxicities:any immune-related AE** (grade any); **pneumonitis** (grade >=3); **colitis** (grade >=3); **thyroiditis (any)** (grade any); **hepatitis** (grade >=3)
- **Efficacy:ORR** 29% (95% CI 21-39); **ORR** 6%; **DoR**

CDKN2A-loss / MTAP-targeting interventions

1. Abemaciclib (CDK4/6 inhibitor, CNS-penetrant) off-label or on a PRMT5 + CDK4/6 combination protocol at first progression — leveraging the 9p21 co-deletion (CDKN2A loss + MTAP loss) synthetic-lethal logic; gated on RB1 functional confirmation

Likelihood of effect: Low for monotherapy (PFS6 ~9.7% recurrent CDKN2A-loss GBM, Wen 2024); the combination case (CDK4/6i + PRMT5i) is preclinical and untested in clinic.

Toxicity burden: Moderate (G≥3 diarrhea characteristic of class, neutropenia, fatigue; at age 66 start at 150 mg BID rather than on-label 200 mg BID)

Counter-productive MoA:Moderate (Pgp efflux at the BBB blunts the CDK4/6 class in CNS (Taal palbociclib failure); abemaciclib was selected for partial CNS penetration but intratumoral concentrations are modest.)

Overall: Recurrence-time pre-position keyed to the patient's exact 9p21 co-deletion biology; monotherapy ceiling is modest and the combination thesis is preclinical, so the rank is honest about the evidence gap.****

Key references: NCT02981940 · PMID 37722087 · PMID 30151703

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
Anvumetostat (AMG 193) monotherapy or + docetaxel (docetaxel arms not relevant here) (basket/biomarker)	small_molecule	1/2	Advanced MTAP-null solid tumors (MTAPESTRY 101); Part 1m is a dedicated glioma cohort	active not recruiting	NCT05094336
S095035 (MAT2A inhibitor) monotherapy or + TNG462 (PRMT5 inhibitor) (basket/biomarker)	small_molecule	1/2	Advanced/metastatic MTAP-deleted solid tumors including IDH-wildtype GBM (explicit cohort)	active not recruiting	NCT06188702
BAY 3713372 (second-generation PRMT5 inhibitor) (basket/biomarker)	small_molecule	1/2	MTAP-deleted advanced solid tumors (first-in-human basket)	recruiting	NCT06914128
TNG462 (peripheral-only PRMT5 inhibitor)	small_molecule	1/2	MTAP-deleted advanced solid tumors EXCLUDING primary CNS	recruiting	NCT05732831
AZD3470 (MTA-cooperative PRMT5 inhibitor)	small_molecule	1/2	Advanced MTAP-deficient solid tumors (PRIMROSE); primary CNS malignancy excluded	recruiting	NCT06130553
ISM3412 (MAT2A inhibitor)	small_molecule	1/2	Locally advanced/metastatic MTAP-null solid tumors; active CNS primary tumor excluded	recruiting	NCT06414460
IDE397 (MAT2A inhibitor) monotherapy and combinations (e.g. + anvumetostat — see NCT05975073)	small_molecule	1	Advanced MTAP-deleted solid tumors (lung, bladder, others — no GBM cohort)	active not recruiting	NCT04794699
Anvumetostat (AMG 193) monotherapy	small_molecule	1	MTAP-null advanced solid tumors (MTAPESTRY 101 phase 1 backbone)	active not recruiting	NCT05094336
MRTX1719 (PRMT5-MTA inhibitor)	small_molecule	1	MTAP-deleted advanced solid tumors (mesothelioma, NSCLC, MPNST, pancreatic — no CNS)	recruiting	NCT05245500
IDE892 (next-generation MAT2A inhibitor)	small_molecule	1	MTAP-deleted advanced solid tumors (NSCLC, GEJ, urothelial, mesothelioma; no GBM cohort listed at start)	recruiting	NCT07277413
Standard TMZ control vs. neratinib + TMZ vs. abemaciclib + TMZ (closed to accrual) vs. CC-115 (closed) vs. QBS10072S	small_molecule	2	Newly diagnosed IDH-wildtype GBM, MGMT unmethylated	recruiting	NCT02977780

TNG908 (brain-penetrant PRMT5 inhibitor) (basket/biomarker, discontinued)	small_molecule	1/2	MTAP-deleted advanced solid tumors including relapsed/refractory GBM (dedicated GBM expansion arm)	terminated	NCT05275478

Evidence in detail

Abemaciclib (CDK4/6 inhibitor, CNS-penetrant) off-label or on a PRMT5 + CDK4/6 combination protocol at first progression — leveraging the 9p21 co-deletion (CDKN2A loss + MTAP loss) synthetic-lethal logic; gated on RB1 functional confirmation

Wen PY et al. *Clinical Cancer Research* 2023

- **Disease context:Newly diagnosed MGMT-unmethylated glioblastoma** (1L (adjuvant, post-RT)) — Adults with newly diagnosed MGMT-unmethylated IDH-wildtype GBM, post-resection or biopsy, completed standard chemoradiation. Bayesian adaptive randomization across abemaciclib, neratinib, CC-115, and control TMZ arms.
- **n:** 73
- **Toxicities:diarrhea** (grade any); **neutropenia** (grade >=3); **fatigue** (grade any)
- **Efficacy:HR_PFS** 0.72 (95% CI 0.49-1.06); **HR_OS**; **HR_PFS** 0.72 (95% CI 0.5-1.02)

Taal W et al. *Journal of Neuro-Oncology* 2018

- **Disease context:Recurrent glioblastoma, RB1-proficient by IHC** (2L+) — Adults with recurrent GBM, KPS >=60, RB1 protein expression confirmed by IHC, <=3 prior relapses. Arm 1 = palbociclib pre-op then adjuvant; Arm 2 = palbociclib without resection.
- **n:** 22
- **Toxicities:neutropenia** (grade >=3); **thrombocytopenia** (grade >=3); **fatigue** (grade any)
- **Efficacy:PFS** 1.2 months; **OS** 3.6 months; **PFS_rate** 5%

Egfr Viii interventions

1. Front-line EGFRvIII / EGFR-amp CAR-T (MGH CARv3-TEAM-E NCT05660369 Arm 2 newly-diagnosed unmethylated; UCSF E-SYNC NCT06186401 Cohort 1; Penn dual-CAR family NCT07209241) — front-line use NOT recommended in this patient

Likelihood of effect: Hypothesis-generating only — Choi 2024 n=3 transient signal with 2/3 progressing inside two months; class signal is single-digit-n single-arm with no OS validation.

Toxicity burden: High (Bagley 2024 100% transient ICANS-like neurotoxicity, Goff 2019 1 treatment-related death at >1e10 cells, hemorrhage-compounded edema risk in this patient)

Counter-productive MoA:High (CRS / ICANS-mediated cerebral edema on a pre-bled brain is the conservative-flagged mechanism-level risk; no phase 1 cohort specifically addresses re-bleed risk under cellular-therapy inflammation.)

Overall: Front-line CAR-T is not recommended in this patient — conservative veto stands on**

hemorrhage-compounded CRS / ICANS edema risk that the published phase 1 cohorts do not address. Recurrence-time conversation re-opens with EGFRvIII positivity and a hemorrhage-aware re-bleed surveillance plan.**

Key references: PMID 38477966 · PMID 38497615 · PMID 30418325 · NCT05660369 · NCT06186401 · NCT07209241

Evidence in detail

Front-line EGFRvIII / EGFR-amp CAR-T (MGH CARv3-TEAM-E NCT05660369 Arm 2 newly-diagnosed unmethylated; UCSF E-SYNC NCT06186401 Cohort 1; Penn dual-CAR family NCT07209241) — front-line use NOT recommended in this patient

Choi BD et al. *N Engl J Med* 2024

- **Disease context:Recurrent EGFR-expressing glioblastoma** (2L+) — 3 adults with recurrent GBM expressing EGFR (not requiring vIII); intraventricular delivery via Ommaya. Single-center MGH.
- **n:** 3
- **Toxicities:any TRAE** (grade >=3): 0% (n=3); **fever** (grade any); **headache** (grade any); **CRS** (grade >=2): 0% (n=3); **ICANS** (grade >=3): 0% (n=3)
- **Efficacy:ORR** 100%; **DoR**; **TRAE_rate** 0%

Bagley SJ et al. *Nature Medicine* 2024

- **Disease context:Recurrent EGFR-amplified glioblastoma** (2L+) — 6 adults with recurrent GBM, EGFR amplification (no vIII required), measurable disease. Intrathecal delivery via Ommaya reservoir.
- **n:** 6
- **Toxicities:neurotoxicity (ICANS-like)** (grade any): 100% (n=6); **neurotoxicity** (grade >=3); **CRS** (grade any); **headache** (grade any): 100% (n=6); **fatigue** (grade any): 100% (n=6)
- **Efficacy:biomarker**; **ORR** 0%; **DoR**

Goff SL et al. *Journal of Immunotherapy* 2019

- **Disease context:Recurrent EGFRvIII-positive glioblastoma** (2L+) — 18 adults with EGFRvIII+ recurrent GBM, KPS >=60; lymphodepletion pre-conditioning before CAR-T infusion. NCI Surgery Branch.
- **n:** 18
- **Toxicities:treatment-related death** (grade 5): (n=18); **CRS** (grade >=3): 0% (n=17); **lymphodepletion-related cytopenias** (grade >=3)
- **Efficacy:ORR** 0%; **OS_rate**; **TRAE_rate**